Graph Polar Equations by Plotting Points

A solution to a polar equation is an ordered pair (r,θ) that satisfies the equation.
The graph of a polar equation is the graph of all solutions to the equation.

$$r = 3\cos\theta$$

1. Graph by plotting points

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$-\pi/6$ ($11\pi/6$)	$-\pi/4$ ($7\pi/4$)	$-\pi/3$ ($5\pi/3$)	$-\pi/2$ ($3\pi/2$)
r	3	$\frac{3\sqrt{3}}{2}$	$\frac{3\sqrt{2}}{2}$	$\frac{3}{2}$	0	$\frac{3\sqrt{3}}{2}$	$\frac{3\sqrt{2}}{2}$	$\frac{3}{2}$	0
$r \approx$	3	2.6	2.1	1.5	0	2.6	2.1	1.5	0

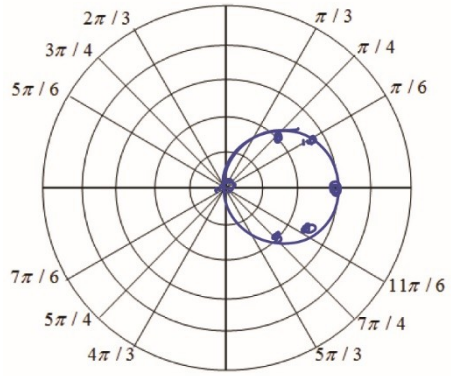
$r = 3\cos 0 = 3$

$r = 3\cos\frac{\pi}{6} = \frac{3\sqrt{3}}{2}$

$r = 3\cos\frac{\pi}{4} = \frac{3\sqrt{2}}{2}$

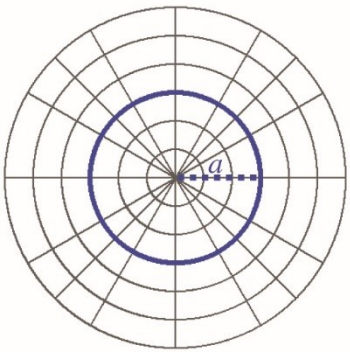
$r = 3\cos\frac{\pi}{3} = \frac{3}{2}$

etc.



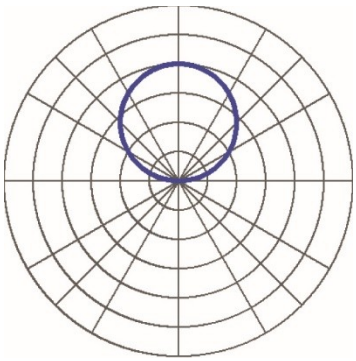
Circles:

$r = a$



Center at the pole, radius $|a|$

$r = a\sin\theta$

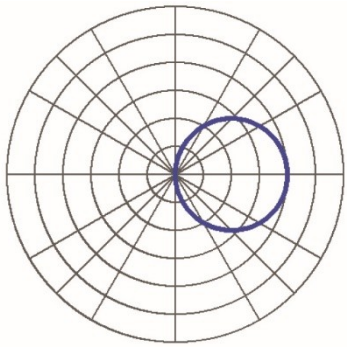


Center on $\theta = \pi/2$, diameter $|a|$

if a is negative, circle is below polar axis.

$r = a\cos\theta$

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Center on polar axis, diameter $|a|$
if a is negative, circle is left of pole.

$r = a\cos\theta$

cos moves circle

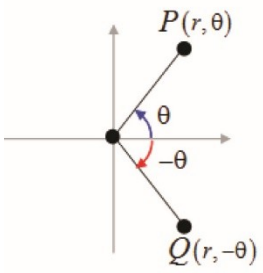
left & right

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Test Polar Equations for Symmetry

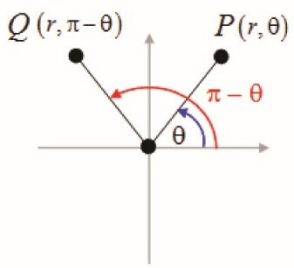
checking for reflections

Symmetry with respect to the polar axis



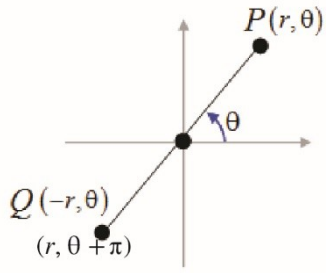
Replace (r,θ) by $(r,-\theta)$

Symmetry with respect to the line $\theta = \frac{\pi}{2}$



Replace (r,θ) by $(r,\pi-\theta)$

Symmetry with respect to the pole



Replace (r,θ) by $(-r,\theta)$

If the given substitution does not give an equivalent equation, the test for symmetry is inconclusive.
If the test for symmetry fails, the graph may still have that symmetry.
Remember: there are an infinite number of representations of a given point.

Another tool: identify the maximum value of $|r|$ if it exists.

The maximum value of $|r|$ gives the distance from the pole of the point or points on a graph that are farthest from the pole. This can be helpful in graphing a polar equation by hand or when setting up a viewing window for a graphing utility.

Determine whether the tests for symmetry given above detect symmetry with respect to

a. The polar axis.

b. The line $\theta = \frac{\pi}{2}$.

c. The pole.

Otherwise, indicate that the test is inconclusive.

2. $r = 8\cos 2\theta$

a. $r = 8\cos 2(-\theta) = 8\cos 2\theta$ ✓ polar axis

b. $r = 8\cos 2(\pi - \theta)$

$= 8\cos(2\pi - 2\theta)$

$= 8\cos 2\theta$ ✓ line $\theta = \frac{\pi}{2}$

c. $-r = 8\cos 2\theta$

$r = -8\cos 2\theta$

inconclusive

3. $r = 4 + 4\sin\theta$

a. $r = 4 + 4\sin(-\theta)$

$r = 4 - 4\sin\theta$ inconclusive

b. $r = 4 + 4\sin(\pi - \theta)$

$r = 4 + 4\sin\theta$ ✓ line $\theta = \frac{\pi}{2}$

c. $-r = 4 + 4\sin\theta$

$r = -4 - 4\sin\theta$ inconclusive

Polar axis $(r,\theta) \to (r,-\theta)$ Pole $(r,\theta) \to (-r,\theta)$

Line $(r,\theta) \to (r,\pi-\theta)$

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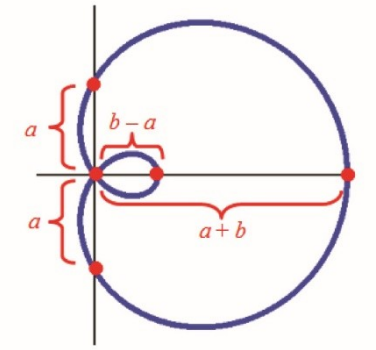
Categorize Polar Equations and Their Graphs

Limaçons $r = a \pm b\cos\theta$ $r = a \pm b\sin\theta$ $(a > 0, b > 0)$

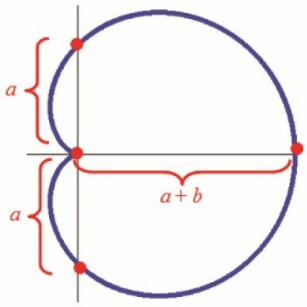
The orientation and symmetry depend on the trigonometric function and the sign of the sine or cosine term.

With respect to the axis $\theta = \frac{\pi}{2}$: $r = a + b\cos\theta$ (right) $r = a - b\cos\theta$ (left)
$r = a + b\sin\theta$ (up) $r = a - b\sin\theta$ (down)

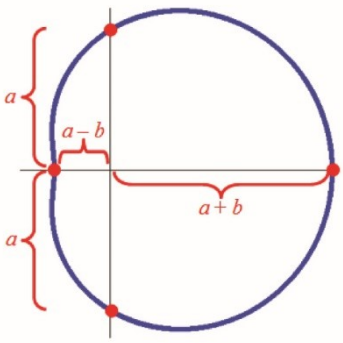
Limaçon with a loop $a < b$ (or $\frac{a}{b} < 1$)



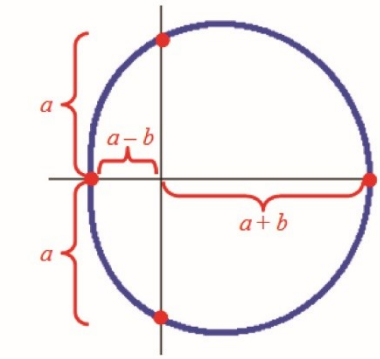
Cardioid $a = b$ (or $\frac{a}{b} = 1$)



Limaçon with a dimple $b < a < 2b$ (or $1 < \frac{a}{b} < 2$)

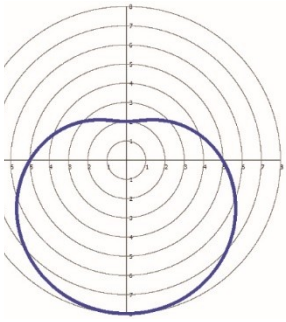


Convex limaçon $a \quad 2b$ (or $\frac{a}{b} \quad 2$)



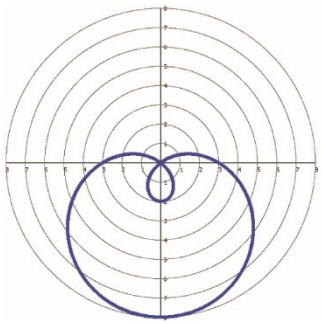
Examples:

$r = 5 - 3\sin\theta$

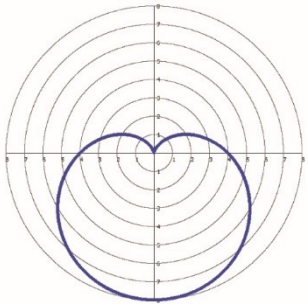


$r = 3 - 5\sin\theta$

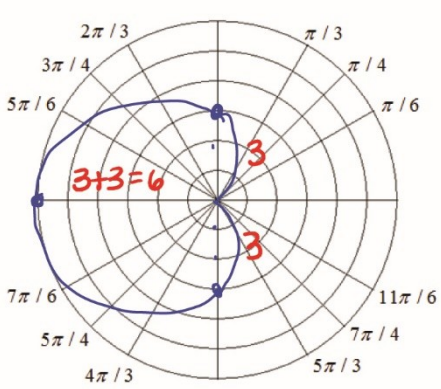
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$r = 4 - 4\sin\theta$



4. Graph $r = 3 - 3\cos\theta$. $r = a - b\cos\theta$, $a = b$, cardioid, left



	at pole
$r = 3 - 3\cos(-\theta)$	$0 = 3 - 3\cos\theta$
$r = 3 - 3\cos\theta$	$-3 = -3\cos\theta$
sym. with respect to polar axis	$1 = \cos\theta$
	$\theta = 0, 2\pi$

max value = 6

$6 = 3 - 3\cos\theta$

$3 = -3\cos\theta$

$-1 = \cos\theta$

$\theta = \pi$

θ	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
r	3	$\frac{9}{2}$	$\frac{6+3\sqrt{2}}{2}$ ≈ 5.1	$\frac{6+3\sqrt{3}}{2}$ ≈ 5.6	6

Roses $r = a\cos n\theta$ $r = a\sin n\theta$ ($|a|$ is length of each petal)

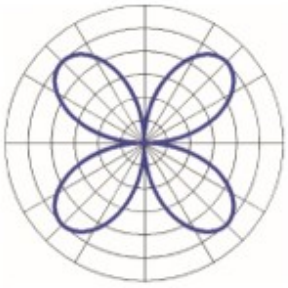
n petals if n is odd, $2n$ petals if n is even To find spacing between petals: $\frac{360°}{\#\text{ of petals}}$ or $\frac{2\pi}{\#\text{ of petals}}$

To find where petal begins: solve $|a| = a\cos n\theta$ or $|a| = a\sin n\theta$

Examples:

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$r = 5\sin 2\theta$ → even → 4 petals



each petal spaced apart $\frac{2\pi}{4} = \frac{\pi}{2}$

max value (petal length) $|5| = 5$

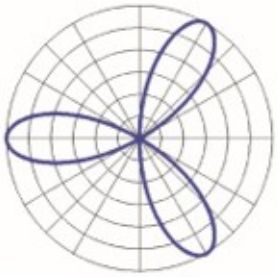
$5 = 5\sin 2\theta$

$1 = \sin 2\theta$

$2\theta = \frac{\pi}{2}$

$\theta = \frac{\pi}{4} \leftarrow$ begin

$r = -6\cos 3\theta$ → odd → 3 petals



each petal spaced apart $\frac{2\pi}{3}$

max value (petal length) $|-6| = 6$

$6 = -6\cos 3\theta$

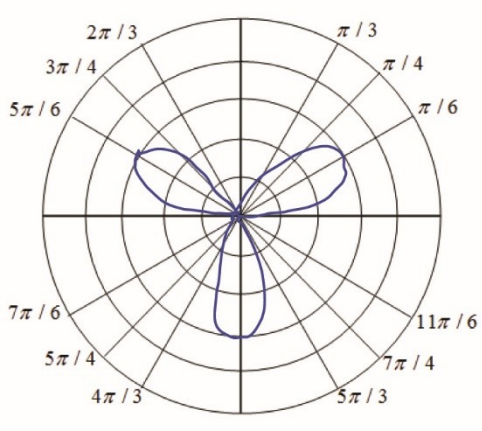
$-1 = \cos 3\theta$

$3\theta = \pi$

$\theta = \frac{\pi}{3} \leftarrow$ begin

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5. Graph $r = 4\sin 3\theta$. 3 petals



at pole

$4\sin 3\theta = 0$

$\sin 3\theta = 0$

$3\theta = 0, \pi, 2\pi, 3\pi, \ldots$

$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \ldots$

spacing: $\frac{2\pi}{3}$

max value

$4 = 4\sin 3\theta$

$1 = \sin 3\theta$

$3\theta = \frac{\pi}{2}$

$\theta = \frac{\pi}{6} \leftarrow$ begin

or

$3\theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \ldots$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6}, \ldots$ petals

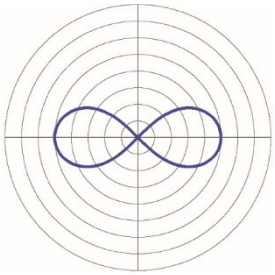
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Lemniscates $r^2 = a^2\cos 2\theta$ $r^2 = a^2\sin 2\theta$ (each "lobe" of length a)

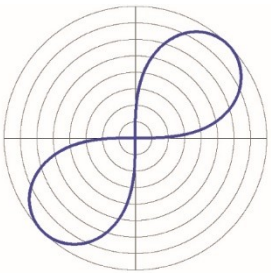
Figure-eight graph symmetric to the polar axis (cosine) or symmetric to the line $\theta = \frac{\pi}{4}$ (sine)

Examples:

$r^2 = 25\cos 2\theta$



$r^2 = 64\sin 2\theta$



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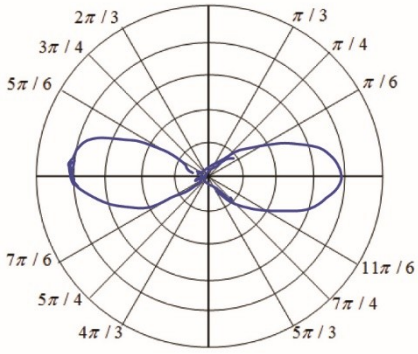
6. Graph $r^2 = 16\cos 2\theta$.

at the pole $16\cos 2\theta = 0$

$\cos 2\theta = 0$

$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$

$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$



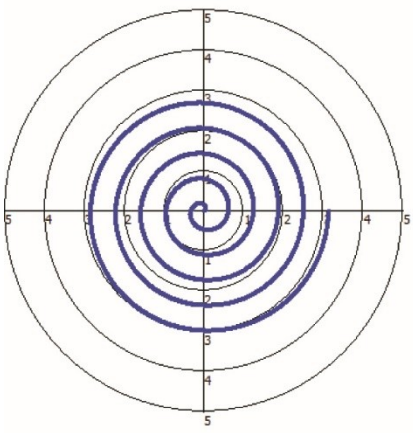
max value

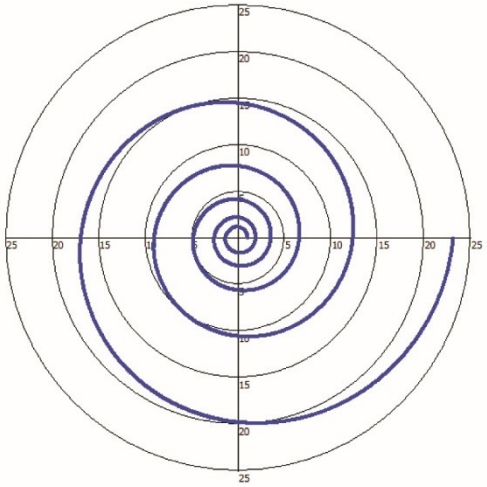
$r = 4$

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Spirals $r = a\theta$ (Archimedean spiral) $r = e^{a\theta}$ (logarithmic spiral)

Examples: $(0 < \theta < 10\pi)$ $r = 0.1\theta$ $r = e^{0.1\theta}$





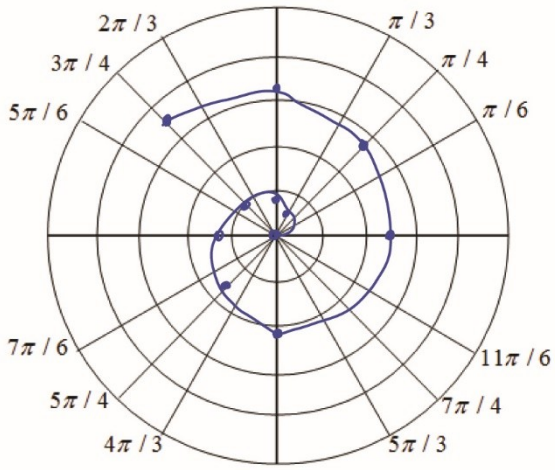
$r = 0.4\theta.$

7. Graph

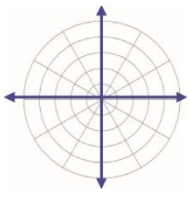
θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{7\pi}{4}$
$r \approx$	0	0.3	0.6	0.9	1.3	1.6	2.2

θ	2π	$\frac{9\pi}{4}$	$\frac{5\pi}{2}$	$\frac{11\pi}{4}$	3π
$r \approx$	2.5	2.8	3.1	3.5	3.8

etc.

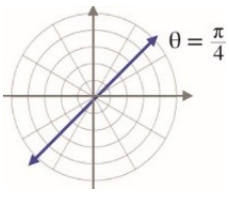


Lines

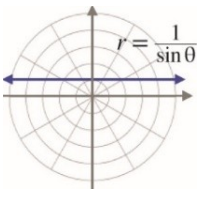


$\theta = 0$ x-axis

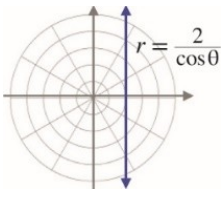
$\theta = \frac{\pi}{2}$ y-axis



$\theta = a$
Line through origin



$r = \frac{a}{\sin\theta}$
Horizontal line
$(y = a)$



$r = \frac{a}{\cos\theta}$
Vertical line
$(x = a)$

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